GetOutSoftware Project Report

By Seneca InnWIL Team Lead with your Heart
Arushi Gopinath
Jayme Laso-Barros
Rojin Saru
Xingyi Chen (Theo)
Zixu Fan (Jeff)

Prepared for Lead With Your Heart Corp.

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Project Objective & Client Information

Executive Summary

The GetOutSoftware consultation was a collaborative initiative between Lead With Your Heart Corp. and the Seneca InnWIL Team Lead with your Heart to assess the viability of a new emergency management software tailored to the needs of the vulnerable and remote communities in Canada. This initiative emerged from a critical recognition that existing evacuation communication systems are often fragmented, inaccessible, and not designed with the lived realities of marginalized populations in mind. Through a blend of structured interviews with emergency professionals and detailed analysis of after-action reports, alongside valuable opportunities for technological innovation, the team aimed to provide recommendations for our client on the most effective path to developing a software that is a net benefit to communities facing disasters.

The project was aimed to bridge the disconnect between emergency responders and software developers by grounding product design in operational challenges, community experiences, and data-driven insights. By identifying deficiencies in communication infrastructure, coordination, and software usability, this report provides a foundation for the design and development of the GetOutSoftware, a platform that prioritizes inclusivity, clarity, and resilience. The findings informed recommendations that centre on developing a system that is intuitive, adaptable, interoperable, and capable of operating under the harsh realities of disaster scenarios, including infrastructure breakdowns and limited internet access.

Brief Overview of the Consultation

The GetOutSoftware consultation was a collaborative initiative led by our client, Lead With Your Heart Corp., in partnership with our team, Seneca InnWIL Team Lead with your Heart, to assess the feasibility of developing a new emergency management tool. This project emerged from the recognition that the current evacuation communication methods are often soft, fragmented, and not tailored to the needs of vulnerable populations. The consultation involved engaging with emergency management professionals and other stakeholders to explore how evacuation plans are currently communicated, what tools are being used and what gaps exist in the process. The goal is to gather insight on the challenges faced during evacuation, and identify the features and functions that an ideal solution should include. This feedback will aid in shaping the design and direction of the GetOutSoftware, ensuring it addresses real-world needs and improves safety, efficiency, and preparedness during emergencies.

Purpose of the Consultation

The purpose of the consultation was to explore the feasibility, design requirements, and real-world applicability of developing a new emergency management software solution called GetOutSoftware, led by Lead With Your Heart Corp. and our team, Seneca InnWIL Team Lead with your Heart. The consultation was aimed to gather insights directly from professionals involved in emergency response, community leadership, and disaster planning, in order to better understand the specific gaps and challenges within current evacuation communication systems.

The consultation was initiated in response to recurring issues observed during large-scale emergencies, such as lack of real-time information, fragmentation of communication between

agencies, absence of trauma-informed planning for vulnerable populations, and the overreliance on complex or inaccessible technologies. These issues disproportionately affect remote and Indigenous communities, families with children, and evacuees with specific medical, legal, or cultural needs.

Through structured interviews and secondary research, the consultation sought to validate the need for a centralized, user-friendly, and culturally responsive software platform that would streamline coordination among responders, host communities, and evacuees. It also aimed to identify the core features such a tool should include, such as offline functionality, customizable alerts, integrated resource tracking, real-time evacuee data, and language accessibility.

Ultimately, the consultation served to bridge the gap between software developers and emergency service practitioners by ensuring that any technological solution proposed is grounded in the lived experiences and operational realities of those who would use it. The insights gained during this consultation directly informed the proposed direction, structure, and priorities for the development of the GetOutSoftware.

Key Findings and Insights

The consultations and interviews revealed several consistent issues across various emergency scenarios. Communication failures were among the most pressing problems, especially in remote regions where internet and mobile networks are unreliable. During events like the High River Flood, and Yellowhead County Wildfires and Floods, local officials were forced to rely on labor intensive methods such as door to door knocking and bullhorns due to system collapse. This not only hindered evacuation efforts but also jeopardized public safety.

Another major insight was the lack of tailored planning for vulnerable populations. Many software systems currently in use do not account for evacuees' medical needs, language barriers, or trauma history (i.e domestic abuse). The consultation also highlighted the fragmentation of existing tools that municipalities often need to operate multiple platforms simultaneously for personnel tracking, mapping, and logistics which increases complexity and the likelihood of miscommunication. Additionally, most platforms do not support offline functionality or integration with Canada's Incident Command System (ICS) which limits their usefulness in real-time disaster response.

Another one of the most critical findings from the consultations, was the consistent lack of technology adoption in rural and remote communities, particularly among Indigenous populations and small municipalities. Interviews with emergency response professionals and community leaders revealed that, in many areas, essential tasks such as evacuee registration, inventory tracking, and personnel management are still conducted manually using pen and paper, spreadsheets, or basic office tools. This reliance on low-tech solutions persists despite the increasing frequency and severity of natural disasters.

Overall, the consultation and interviews underscored that the issue is not simply a lack of access to software, but a broader problem of fit, usability, and trust. There is a strong need for intuitive, modular, and community adapted technology solutions that do not require constant internet connectivity or extensive training.

Summary of Recommendations

A critical recommendation emerging from the consultation is the need for robust and fully integrated ICS form functionality within the GetOutSoftware platform. The Incident Command System, or ICS, is the foundational structure used across Canada to organize and coordinate emergency response operations. It provides clarity in command, role designation, and standardized documentation that ensures all agencies and responders can work together effectively. Despite its importance, many existing emergency management software systems do not adequately support ICS documentation. As a result, responders often rely on printed forms, static PDF documents, or separate systems to handle critical information. This leads to administrative overload, delays in coordination, and greater risk of mistakes, especially when operations are taking place under pressure.

To address this, ICS forms must be deeply integrated into the core design of GetOutSoftware. The platform should allow users to access, complete, and manage the full range of forms used in Canadian emergency operations. These include documents related to incident briefings, organizational assignments, communication plans, medical planning, and activity logging. The forms must be interactive and editable in real time, with automatic population of fields wherever data is already available within the system. This might include personnel assignments, 45 geographic data, and operational resources. By reducing the need for repeated manual entry, the software would save time and reduce the chance of errors.

One of the most important features of ICS form integration is the ability to function in offline environments. Many emergencies occur in rural or remote areas where internet access is unreliable or completely unavailable. The software must allow users to download blank forms to their devices, complete them in the field without a connection, and then sync the data back to the central system once connectivity is restored. The system should also allow printing of forms when digital access is not possible, making sure that field teams are never left without the tools they need to record and communicate vital information. Location tracking, such as GPS stamping for activity logs, can help with accountability and situational tracking.

Form validation is another important feature. The software should ensure that required fields are completed properly and that information is entered in a consistent and clear format. Once forms are complete, they should be easy to export into accessible formats like PDF or Excel, allowing for sharing, archiving, or cross-jurisdictional coordination.

By embedding ICS form capabilities directly into GetOutSoftware, the platform will significantly reduce administrative burden and improve clarity across response teams. This feature will also help eliminate the use of parallel systems, making emergency documentation more efficient, reliable, and usable. It will support better coordination, faster decision making, and stronger accountability, all of which are essential in high stakes emergency scenarios.

Scope of Work (Market Research, Competitor Analysis)

The scope of this project is focused on identifying and evaluating the key features necessary to develop an effective, inclusive, and scalable emergency management software tool called GetOutSoftware. This work was undertaken in collaboration with Lead With Your Heart Corp. and our team, Seneca InnWIL Team Lead with your Heart, aimed to support our client's goal of creating a communication and coordination platform for use during evacuation events.

To achieve this, the team conducted comprehensive market research and competitor analysis to better understand the current landscape of emergency management solutions. This involved identifying the limitations of existing software platforms such as Everbridge, DisasterLAN (DLAN), D4H, GSAR (Ground Search and Rescue), Juvare, Navigate360, etc.

In addition to market analysis, this scope included a detailed secondary research on past Canadian disasters through after-action reports. These documents revealed patterns in system failures, communication gaps, and resource bottlenecks, as well as best practices that can be replicated or improved upon.

Another key component of the scope was the direct engagement with industry professionals and partners of our client through interviews. These conversations provided real-world insights into software usability, communication challenges, cultural considerations, and the operational realities faced during crises.

Based on the findings from these efforts, the project also involves ideation and solution development. This included brainstorming new technology features, conducting a SWOT and 4P's analysis, and building a business model canvas to guide potential implementation strategies.

Overall, the scope of work combined research, analysis, consultation, and design planning to deliver actionable recommendations for the development of the GetOutSoftware. The outcomes are intended to inform software development decisions that prioritize simplicity, accessibility, and safety in emergency management.

Methodology Used for Research

The research methodology for this project combined qualitative and document-based approaches to gain a comprehensive understanding of the challenges and opportunities in the current emergency management practices. Specifically, two key methods were used, the analysis of after-action reports and the conduction of stakeholder interviews.

After-action reports (AARs) were selected by the recommendation of our client, from large-scale emergency events across Canada to uncover patterns in response effectiveness, communication breakdowns, and coordination strategies. These reports are formal documents prepared after emergencies to evaluate what worked, what did not, and how future responses can be improved. The team analyzed several AARs, including those from the 2013 High River Flood, the 2016 Wood Buffalo Wildfire, the 2021 Lytton Wildfire, and the 2023 Yellowhead County Wildfires and Floods. These case studies offered crucial insight into the recurring issues such as communication infrastructure failure, reliance on manual systems, lack of centralized data sharing, and the absence of trauma-informed strategies in evacuation planning. The reports also highlighted innovative responses and best practices, such as community call-in lines, the use of social media during infrastructure collapse, and the success of amateur radio stations.

In addition to document analysis, the team conducted qualitative interviews with emergency management professionals, who have direct experience with evacuation planning, logistics coordination, public communications, and crisis software tools. These individuals were selected by our client because of their frontline experience and familiarity with both the operational and technological aspects of disaster response. The interviews explored what tools are currently in use, which features are most and least effective, the usability and accessibility of software solutions, and the social and cultural barriers faced during evacuations. Interviewees also

shared their frustrations with complex systems and emphasized the need for real-time data, offline functionality, intuitive interfaces, and culturally appropriate communication methods.

The integration of these two research methods allowed the team to develop a strong foundation for developing the GetOutSoftware. The after-action reports grounded the research in well documented, evidence based analysis, while the interviews provided human-centred perspectives that revealed gaps not always visible in formal documents. Together, they inform a realistic, practical, and empathetic approach to designing an emergency management tool tailored to the diverse needs of Canadian communities or particularly rural Alberta.

Research & Secondary Analysis

2013 High River Flood After Action Report

The emergency response to the June 2013 flood in the Town of High River was a case study in crisis management under extreme pressure. This was Alberta's most devastating natural disaster to date, and High River was the hardest-hit community. The flood's scale was far beyond what the Town's infrastructure and emergency planning had anticipated. Despite these overwhelming circumstances, the emergency response was effective in several critical areas, while also revealing major gaps in preparedness, coordination, and infrastructure resilience.

The Emergency Operations Centre (EOC) was activated early on the morning of June 20, 2013, shortly after Environment Canada upgraded its forecast. The EOC was operational by 5:30 a.m., and municipal staff began immediate preparations for response. This rapid activation demonstrated the Town's readiness to act, despite limited resources. As the floodwaters rose far beyond expected levels, the Town's leadership made a pivotal and life-saving decision to shift the focus from infrastructure protection to rescue operations. This change in priorities likely prevented additional loss of life, especially given that the actual flood flow reached up to 2,000 cubic metres per second, compared to the 750 cubic metres per second (cms) level for which protections were designed.

Rescue efforts were carried out using every available means. Firefighters, operations staff, and contractors employed amphibious vehicles, airboats, combines, and rock trucks to reach stranded residents. Though formal rescue resources were limited, the creative deployment of unconventional equipment was highly effective. These efforts were further bolstered by the arrival of the Canadian Armed Forces and the RCMP. Over the course of the emergency, more than 1,000 responders and operations personnel participated in the field, including volunteers, NGOs, and municipal and provincial agencies.

Evacuation operations were largely successful, with approximately 13,400 residents evacuated and 6,300 structures cleared. Reception Centres established in nearby communities such as Blackie, Nanton, and Okotoks provided shelter and support for over 1,000 evacuees. These centres were able to maintain food service, health access, and support networks even though evacuees remained for up to two weeks in some cases. The Town's decision to delegate its evacuation and emergency social services responsibilities to neighboring municipalities proved essential to sustaining the response.

Another success story was the rescue and care of animals. While the Municipal Emergency Management Plan did not include provisions for animal evacuation, over 1,100 pets were rescued and an additional 400 were supported in their homes. This was possible through coordination between Bylaw Services and organizations such as the Calgary Humane Society,

ASPCA, and RCMP. The operation transitioned from informal paper tracking to more structured procedures as additional personnel arrived.

However, the response faced significant weaknesses that impaired coordination and public safety. The most critical failure occurred in communication infrastructure. By 11:00 a.m. on June 20, the town had lost internet access, landline service, and the mobile network. This collapse left the Town unable to effectively notify the public, issue warnings, or coordinate internally. The Town's website crashed due to high traffic, and the Primary Rate Interface for TELUS lines failed, cutting voice communication at a vital moment. As a result, the Town had to rely on resource-intensive alternatives such as door knocking and bullhorn announcements, which could not reach all areas of the community.

Preparedness planning also fell short of the actual disaster. The Town's flood mitigation measures were designed to withstand a 1-in-100-year flood (750 cms), and planning protocols assumed an eight-hour warning period. In reality, conditions deteriorated much faster, and the river peaked at more than twice the anticipated volume. Consequently, sandbagging and berm construction were overwhelmed before they could be fully deployed. This gap between expected and actual conditions revealed the need for revised flood modelling, updated risk thresholds, and more aggressive mitigation planning.

Human resource limitations further constrained the Town's response. As a small municipality, High River was not equipped to staff both the EOC and field operations for more than a few days. Fatigue, unclear staff roles, and the loss of personal phones and computers among staff members caused delays and confusion. Additionally, the Town lacked a centralized accountability system for staff check-in and safety confirmation. These limitations created operational inefficiencies and made it difficult to maintain sustained command and control.

Despite these shortcomings, the measurable outcomes of the emergency response were relatively positive. Only two fatalities occurred despite the vast scale of the disaster. Most residents were evacuated or rescued in time, and no widespread failure in emergency care or sheltering occurred. Pet rescue, though unplanned, was remarkably successful. The Town also transitioned into recovery quickly, requesting a Provincial State of Emergency on June 27, enabling additional resources and coordination support. Re-entry planning began soon after, and recovery operations were underway before the end of June.

Lessons learned from the flood have since been integrated into the Town's emergency management framework. The Emergency Management Bylaw was revised, and a number of plans were completed or updated, including the Crisis Communications Plan, Business Continuity Plan, Emergency Social Services Plan, and a Public Education Program. A new flood response plan is in development, and planning now accounts for flood levels up to 1,850 cms plus one metre of freeboard. Training programs have been expanded, with over 200 staff trained in new procedures. A Critical Infrastructure Protection Plan and staff care strategies were also added.

In conclusion, the Town of High River's emergency response to the June 2013 flood can be assessed as moderately effective under extreme circumstances. Although unprepared for the scale and speed of the disaster, municipal leadership demonstrated flexibility, courage, and a strong commitment to protecting residents. The community's response was strengthened by inter-agency collaboration, community volunteerism, and decisive actions taken under pressure. The failures in infrastructure and planning were significant but have led to meaningful improvements in the Town's emergency preparedness. The High River flood response stands as an example of adaptive emergency management in the face of overwhelming odds.

Communication Tools Used

Effective communication during a disaster is crucial for ensuring public safety, coordinating emergency response efforts, and maintaining public trust. In the case of the June 2013 flood in High River, the Town employed a wide array of communication tools. These tools were used to distribute warnings, evacuation notices, situation updates, and recovery information. However, the disaster exposed serious vulnerabilities in the Town's communication infrastructure, which greatly affected the overall effectiveness of the emergency response.

Before the flood reached its peak, the Town of High River relied heavily on electronic communication platforms to inform its residents. The Town's official website (www.highriver.ca) was a central information hub, where bulletins, emergency declarations, and flood forecasts were posted. However, as the flood crisis escalated and website traffic surged, the server infrastructure proved inadequate. The site crashed due to high demand on June 20, becoming intermittently accessible over the following days. This severely limited residents' ability to access official information during the most critical hours of the emergency.

To supplement the website, the Town used social media, which played a central role in public engagement and real-time communication. The Facebook page "TownofHighRiver" was used to post flood alerts, evacuation instructions, and government announcements. The page saw a dramatic increase in followers, around 4,500 new followers during the flood. Similarly, the Twitter account "@townofhighriver" gained approximately 3,000 new followers during this time. Twitter was particularly useful for short, urgent updates and was used extensively by residents and media outlets to share information using hashtags such as #ABflood.

The Town also employed mass email communications, including updates sent to media outlets, local radio DJ booths, and subscribers of the Town Crier e-newsletter. The newsletter, which before the flood had 1,648 subscribers, grew by about 1,350 more during the emergency. Between June 20 and August 6, a total of 27 email updates were issued. However, as the flood knocked out power and internet services across large parts of the community, digital email became unreliable for many residents. To address this, the Town printed hard copies of the Town Crier and distributed them by hand, an important example of reverting to analog methods when digital infrastructure failed.

In addition to online platforms, the Town used radio broadcasting as a critical communication tool. The mayor recorded messages that were aired on all three local radio stations. During the height of the crisis, these messages were played 4 to 7 times per day following the news. As recovery progressed and people returned to their homes, the frequency was reduced to twice a week. Radio proved especially valuable because it remained accessible even when other forms of communication failed, provided residents had battery-powered or car radios.

When both telecommunications and the internet collapsed, specifically around 11:00 a.m. on June 20, the town was forced to rely on more manual forms of communication. Bullhorns were used in the streets to deliver evacuation orders and important updates. Town staff and emergency responders physically went door-to-door to notify residents. These methods, while labor-intensive, were critical during the telecom blackout. However, they were limited in geographic reach and could not keep pace with the rapid spread of floodwaters.

Another vital but underutilized tool was the Amateur Radio Emergency Service (ARES). On June 21, ARES was used to re-establish communication between the Emergency Operations Centre and the reception centres in Blackie and Nanton. Amateur radio filled the gap left by the failure of both landlines and cellular networks. Although used only briefly, its success

underscored the importance of maintaining low-tech communication backups during emergencies.

Mobile communication tools such as text messaging and cellular phones were in use at the beginning of the emergency but were soon rendered ineffective. TELUS phone lines failed as the floodwaters disrupted the Primary Rate Interface (PRI), and the mobile network shut down around 11:00 a.m. on June 20. This total failure of telecom services was one of the most significant communication breakdowns during the crisis. It not only prevented the Town from issuing updates to residents but also made it nearly impossible to coordinate emergency services, evacuations, and rescues effectively.

In response to this collapse, the Town utilized any remaining channels available. One such method was the Alberta Emergency Alert System, which broadcasted emergency declarations and alerts at a provincial level. While it served its purpose in the early stages, the system's reach was limited once local infrastructure went down and many residents had already evacuated or lost access to their devices.

Following the peak of the flood, communication gradually improved. Intermittent internet access returned after June 21, and the mobile network was restored later that day, helping to reestablish some level of coordination and public messaging. The Town resumed publishing its digital newsletters, reopened its website, and increased its social media activity. Information about recovery efforts, re-entry procedures, and volunteer opportunities was distributed through these revived channels.

The breakdown in communication revealed several critical lessons. First, relying solely on digital systems without backup options is a major risk. The Town's website, social media, and phone systems all failed during the emergency, demonstrating the need for redundant systems such as satellite phones, amateur radios, and printed materials. Second, mass communication must be supported by ground-level efforts such as door knocking and bullhorns, particularly when residents are dispersed, without access to electricity or internet, or when infrastructure has failed. Third, partnerships with local media, especially radio stations, are essential and should be formalized in emergency plans.

In conclusion, while High River employed a wide range of communication tools during the 2013 flood, including websites, email, social media, emergency alerts, radios, bullhorns, and printed newsletters, the effectiveness of these tools was uneven. Initial communication efforts were strong but quickly overwhelmed by infrastructure failures. The Town adapted by turning to analog and low-tech methods, but the experience made clear the need for more robust, layered communication planning in future emergencies. Since the flood, the Town has taken steps to develop a more comprehensive Crisis Communications Plan that incorporates these lessons and prioritizes resilience in communication systems.

2023 Yellowhead County Wildfires and Floods After Action Report

Yellowhead County's emergency response in 2023 was both remarkable and strained. On the positive side, the County demonstrated a strong foundational framework for managing emergencies. It had a Municipal Emergency Plan that was fully compliant with provincial regulations and had already established relationships with Alberta Emergency Management Agency (AEMA), Alberta Wildfire, and surrounding municipalities. Leadership alignment among the Mayor, Council, CAO, and Fire Chief enabled swift and coherent decision-making during the unfolding crisis. One of the most effective actions taken was the early request for external assistance. Recognizing the scale of the disaster early on, the County activated mutual aid,

bringing in critical support from the North Central Incident Management Team and Canada Task Force 2. This proactive approach ensured that Yellowhead County was not left to face the disaster in isolation.

However, these strengths were met with significant challenges. Despite having a solid emergency plan, it was reportedly over 800 pages long and difficult for many staff to use effectively, particularly those new to their roles or lacking specific training. While the plan was accessible, many were unfamiliar with its contents, and there was no clear operationalization at the departmental level. Business continuity planning was inconsistent. Only a small portion of staff surveyed knew whether their departments had continuity plans in place. During evacuations, staff were uncertain about their responsibilities, authority, and expectations, leading to confusion and inconsistency in execution.

Operationally, the County's Emergency Coordination Centre adapted quickly and managed a seamless transition from wildfire response to flood response as weather conditions changed dramatically. This flexibility was critical, especially as the event stretched over 63 days. However, the ECC struggled with space, structure, and clarity of roles. Because of resource constraints, the ECC often served dual functions as both a coordination centre and an incident command post, which blurred lines of authority and coordination. Compounding this was the limited staffing capacity. With only 14 full-time firefighters and a fluctuating volunteer base, Yellowhead County's response force was simply not large enough to handle a crisis of this magnitude without significant outside help. Many volunteers worked fewer than four shifts during the event, highlighting the vulnerability of relying on a volunteer-based model in prolonged emergencies.

Communication efforts showed both promise and shortfall. The establishment of a community call-in line was highly effective and widely used by residents from Yellowhead County and surrounding areas. The County also utilized Alberta Emergency Alert, social media, and its official website to broadcast critical information. Yet, despite this multipronged approach, communication often lacked consistency and clarity. Residents were frequently confused about evacuation orders, fire proximity, and the severity of the threat. Approximately one-third of survey respondents reported ignoring evacuation orders, citing reasons such as concern for livestock, unclear risk, and lack of trust in the information provided. Mapping tools provided by Alberta Wildfire were considered outdated and difficult to interpret. This lack of accessible and clear situational awareness made it difficult for residents to make informed decisions and for responders to maintain public trust.

At the interagency level, coordination was generally strong in spirit but weak in structure. AEMA field officers offered valuable support, even though they were stretched thin across multiple jurisdictions. The Alberta First Responder Radio Communications System functioned effectively but suffered from inconsistent naming conventions that hindered real-time coordination across agencies. The division between Alberta Wildfire's green and white zones also complicated efforts. These jurisdictional distinctions were not always clear to responders or residents, leading to confusion about responsibilities and delayed decision-making. Furthermore, there was no formalized unified command structure between Yellowhead County and Alberta Wildfire. This gap became more apparent when external teams from British Columbia and other regions arrived without established protocols for coordination, leading to duplication and misunderstandings.

In conclusion, Yellowhead County's emergency response to the 2023 wildfires and flooding was defined by resilience, adaptability, and proactive leadership. However, the response was also hampered by systemic limitations in staffing, operational clarity, communication, and interagency coordination. The County did an exceptional job under the circumstances, but future

preparedness will require substantial improvements in training, resource planning, communication systems, and inter-agency structures to ensure that the community is better equipped to face events of similar or greater magnitude.

Communication Tools Used

Yellowhead County employed a variety of communication tools during the emergency, each with differing levels of effectiveness and accessibility. One of the most critical systems used was the Alberta Emergency Alert (AEA), which delivered real-time alerts to cell phones. These alerts were the primary official mechanism for notifying residents of evacuation orders, fire activity, and other urgent information. According to the community survey, 78 percent of respondents received these alerts, indicating that the system had broad reach and was relied upon by a large segment of the population. However, while AEA alerts were timely, they were sometimes issued without sufficient context or follow-up communication, which left residents unclear on specific actions they were expected to take.

The Yellowhead County website served as another major communication platform. Residents frequently turned to the site for updated information on evacuation zones, emergency reception centres, fire status, and general safety guidelines. The website was widely used, with 50 percent of survey respondents citing it as a top information source. The county made considerable efforts to keep the content updated, and while the site was generally considered reliable, its effectiveness still depended on whether residents had access to the internet and knew how to navigate to the right resources during a high-stress event.

To supplement static online updates, the County also implemented a dedicated community information call line, which became one of the most positively received tools of the entire response. This call-in line allowed residents to speak directly with trained staff and get accurate, location-specific information about evacuation orders, fire movement, reception centre locations, and more. Its usage extended beyond Yellowhead County to residents in Edson and Parkland County as well. This line was credited with calming anxiety, resolving confusion, and offering a personalized communication channel at a time when trust in mass alerts and social media was wavering. The report specifically highlights this phone line as a best practice that should be maintained and expanded in future events.

Social media platforms, especially Facebook and Twitter, played an important but complicated role in emergency communication. About 70 percent of residents used social media to follow updates, often in real time. While this allowed rapid distribution of information, the nature of social media also made it vulnerable to misinformation, contradictory updates, and emotional responses that clouded clarity. During moments when evacuation zones shifted or multiple fires converged, posts on social media sometimes conflicted with official information, contributing to public confusion. Additionally, some residents relied on word of mouth—friends, family, and neighbours—for information, which, while helpful in some contexts, also carried the risk of spreading unverified or outdated information. Approximately 45 percent of survey respondents reported relying on this informal communication network.

For coordination among emergency personnel, the Alberta First Responder Radio Communication System (AFRRCS) was used. This system is designed to ensure interoperable radio communications between various emergency agencies. It provided a decentralized, reliable backbone for voice communication across departments and jurisdictions. However, its implementation during the 2023 event was hampered by a lack of standardized naming conventions. Different agencies had renamed AFRRCS talk groups on their own systems, which made it difficult to ensure everyone was tuned to the correct channels during joint operations.

This caused delays and inefficiencies during high-pressure moments, especially when new teams arrived from other regions or provinces.

One of the more significant gaps in the communication strategy related to public-facing mapping tools. Alberta Wildfire's mapping platform, while technically available online, was difficult for residents to interpret. It did not offer address-specific overlays or intuitive navigation. As a result, many residents struggled to understand how close the fires were to their homes or how to interpret fire behavior. At one point during the crisis, a team from British Columbia introduced a more sophisticated mapping system that allowed better situational awareness, including fire perimeters and equipment locations. This system was highly praised by local residents and staff, but unfortunately, it was removed when the BC team returned to their province. The lack of a permanent, modernized, and user-friendly mapping solution was repeatedly cited as a problem by both the public and County staff.

The ECC (Emergency Coordination Centre) also used internal briefings and interdepartmental updates to maintain coordination among staff. These communications were sometimes hindered by overlapping terminology and unclear roles. For example, the terms Emergency Coordination Centre, Incident Command Post, Emergency Communications Centre, and Emergency Call Centre were all used inconsistently, which created misunderstandings not just within the County staff, but also among partner agencies and residents.

In summary, Yellowhead County used a broad set of communication tools that covered both mass notification and personalized communication, internal coordination, and public engagement. The effectiveness of these tools varied significantly. The Alberta Emergency Alert and community information call line proved highly valuable, while traditional tools like websites and social media were moderately effective but needed support from clearer, real-time mapping and more consistent messaging. The radio system for responders was functional but undermined by a lack of standardization. Ultimately, while the County demonstrated creativity and adaptability in deploying these tools under pressure, the experience underscored the need for more integrated, modernized, and consistent communication systems in future disaster scenarios.

Lytton Wildfire

Location: Lytton First Nation → Merritt & Kamloops
Context: Evacuees were sent to neighboring host towns that lacked real-time medical data or a trauma-informed strategy after the town was devastated by the Lytton wildfire in 2021.
Problem: The receiving communities were unaware of who had mental health disorders, diabetes, or mobility impairments. There was no centralized follow-up, and the youth were dispersed across the hotels.

Consequences:
Without the staff's knowledge, some people ran out of necessary medication. Officials from the host municipality eventually acknowledged that they were "operating in the dark."

What *GetOutSoftware* Could Have Done:
Refill reminders and a flag for urgent medical requirements were sent out beforehand. GetOutSoftware could have enabled shelters to assign caseworkers and organize families more carefully.

May 2016 Wood Buffalo Wildfire Post-Incident Assessment Report Summary

The May 2016 Wood Buffalo Wildfire (AKA Horse River Wildfire or the Wildfire) was first witnessed on May 1 during the afternoon. Prior to the Wildfire, hazard conditions within the Regional Municipality of Wood Buffalo (AKA the Region) were at “unprecedented” levels, which would soon lead to one of Canada’s worst wildland-urban interface fires. Initially, the Wildfire was roughly two hectares large and located 7 KM outside of the Urban Service Area of Fort McMurray. Due to high winds, it continued to grow and within two days nearly 88,000 people had to be evacuated from the Region. After five weeks, 589,000 hectares of land was affected and 1,958 structures were damaged/lost. The Insurance Bureau of Canada estimated $8.9 billion in damages, making this the costliest insured natural disaster in Canadian history.

At the time, the Region had a traditional population of 81,948 and shadow-population of 43,084, with at least 60% of residents aged between 20-55 years old. While there were no casualties from the Wildfire directly, two people were unfortunately killed in a motor vehicle accident during evacuation. Approximately 100 residents of the Region told their perspective of the incident, while 5,300 others answered a relevant online survey. During this time, 17 other active wildfires were simultaneously happening elsewhere in Alberta, though they all paled in comparison to the one going on in Wood Buffalo. As an example of the sheer scale of this Wildfire, nearly 2,900 firefighters in total were deployed to combat it.

On a municipal level, accountability and responsibility for such disasters within the Region falls upon the Regional Emergency Services Department. They are responsible for the Region’s emergency management by activating the Regional Emergency Operations Centre (REOC) whenever an emergency incident occurs. On a provincial level, the Government of Alberta utilized the Incident Command System (ICS) to help coordinate all operations during an emergency situation. The purpose is to provide for the division of labour and overall safety at the incident level. It also ensures that any work at the incident is done efficiently and effectively.

The Government of Alberta has successfully applied several lessons learned from previous wildfires toward this incident. In addition, KGMP LLP, the writers of this Post-Incident Assessment Report have identified new lessons as a result of the Wildfire and made several recommendations to help grow Alberta’s emergency programs in regards to prevention, preparedness, response, stabilization, and recovery. For details on these lessons and recommendations, please refer to Appendix B.

<u>Key Takeaways from Case Studies</u>

Our review of four major Canadian disaster responses — the 2013 High River Flood, the 2023 Yellowhead County Wildfires and Floods, the 2021 Lytton Wildfire, and the 2016 Wood Buffalo Wildfire revealed consistent patterns in both effective practices and critical shortcomings in disaster management.

What Worked Well:

- Rapid mobilization and evacuation under extreme pressure, often saving lives despite infrastructure failures.
- Strong leadership and inter-agency cooperation once resources were deployed, enabling creative problem-solving.

- Use of available tools and networks like radio, mapping, local knowledge to partially maintain situational awareness.

What Didn't Work:

- Information blind spots: Responders frequently lacked up-to-date details on evacuees' locations, needs, and medical requirements, resulting in preventable risks.
- Communication failures: Cellular outages, fragmented alert channels, and inconsistent terminology created confusion both for the public and between agencies.
- Disjointed recordkeeping: Vital ICS operational forms, maps, and status updates were often scattered across formats and locations, slowing coordination.
- Under-served vulnerable populations: Those with chronic illnesses, mobility challenges, or mental health needs were often missed in the chaos of relocation.

Taken together, these lessons highlight the critical need for integrated, reliable, and accessible systems to improve communication, situational awareness, and coordination during emergencies. They provide a clear framework for guiding the decisions in the following sections, making sure that future solutions address both operational gaps and the needs of vulnerable populations.

Industry Insights & Technology Gaps

To understand the industry and assess the technology gaps, we've looked into several articles, research papers, online forums, and conducted 5 interviews with people from different parts of the industry (interview notes can be found in Appendix A). Upon analyzing the results, we have identified 4 common issues.

Complexity

Many crisis management tools, such as Incident Action Plan (IAP) Software and D4H, are commonly used during emergencies. However, these systems are often too complex for simple incidents or high-stress environments. Even with modular pricing models like that of D4H, organizations often end up paying for unnecessary features. Similarly, GIS-based tools like WebEOC and H2Safety can be helpful but aren't always suitable for every scenario—particularly because their map-centered approach may not work for all types of incidents. These tools also tend to be overly complicated.

In distribution centers, solutions from companies like SAP are often used for inventory management. Yet, they too come with a surplus of features that are not always needed. The result is that these tools require extensive training, consume time and resources, and can even become an obstacle during emergencies. In some cases, a special support team must be deployed on site, which slows down operations.

Fragmentation

Because no single tool can handle all emergency management tasks, organizations often rely on multiple platforms. This leads to fragmentation and interoperability issues, making it difficult for data to flow smoothly between systems. Some teams customize collaborative tools like Microsoft Teams, Slack, Jira, or Confluence to manage emergencies. While these offer flexibility, they can contribute to even more fragmentation when used alongside other systems. Data

constantly moving between platforms also creates a larger attack surface, raising security and privacy concerns.

Slow/Outdated Adoption of Communication Technology

Despite advances in technology, modern tools are still not widely adopted in disaster management. Many organizations remain reliant on manual processes such as spreadsheets or even pen and paper. Communication often happens through landlines, mobile phones, radios, or SMS. In areas where these are unavailable, communication either falls back to face-to-face interaction—or disappears entirely. There is also a general mistrust or discomfort with new technologies among many in the industry, which contributes to the slow adoption rate.

Inefficient Alert System

In some areas, emergency alerts are still delivered door-to-door, especially where internet access is poor. More commonly, alerts go out via mobile phones, SMS, email, or social media. However, many people turn to social media during emergencies, where misinformation spreads quickly. Social platforms also don't prioritize emergency alerts, so important information can be easily missed. Many evacuees also don’t know what to do or what to bring when evacuating, and alerts often fail to provide this information. This lack of clarity causes mental stress, leading to poor decision-making.

Other Issues

Beyond the four common issues above, several other critical issues impact emergency management systems:

- **Customizability**: Existing tools are often built with outdated, inflexible technologies that can’t be tailored to different incidents.
- **Data Accuracy**: Platforms often depend on manual data entry or delayed syncing, leading to outdated or inaccurate information during fast-moving disasters.
- **Language and Cultural Barriers**: Most platforms lack support for Indigenous languages or culturally appropriate communication methods, creating mistrust and confusion among some community members.
- **Licensing Dependency**: Many tools require ongoing, expensive licenses or vendor support, which creates long-term dependency and limits community adaptability, especially as disaster frequency increases.

Recommendations

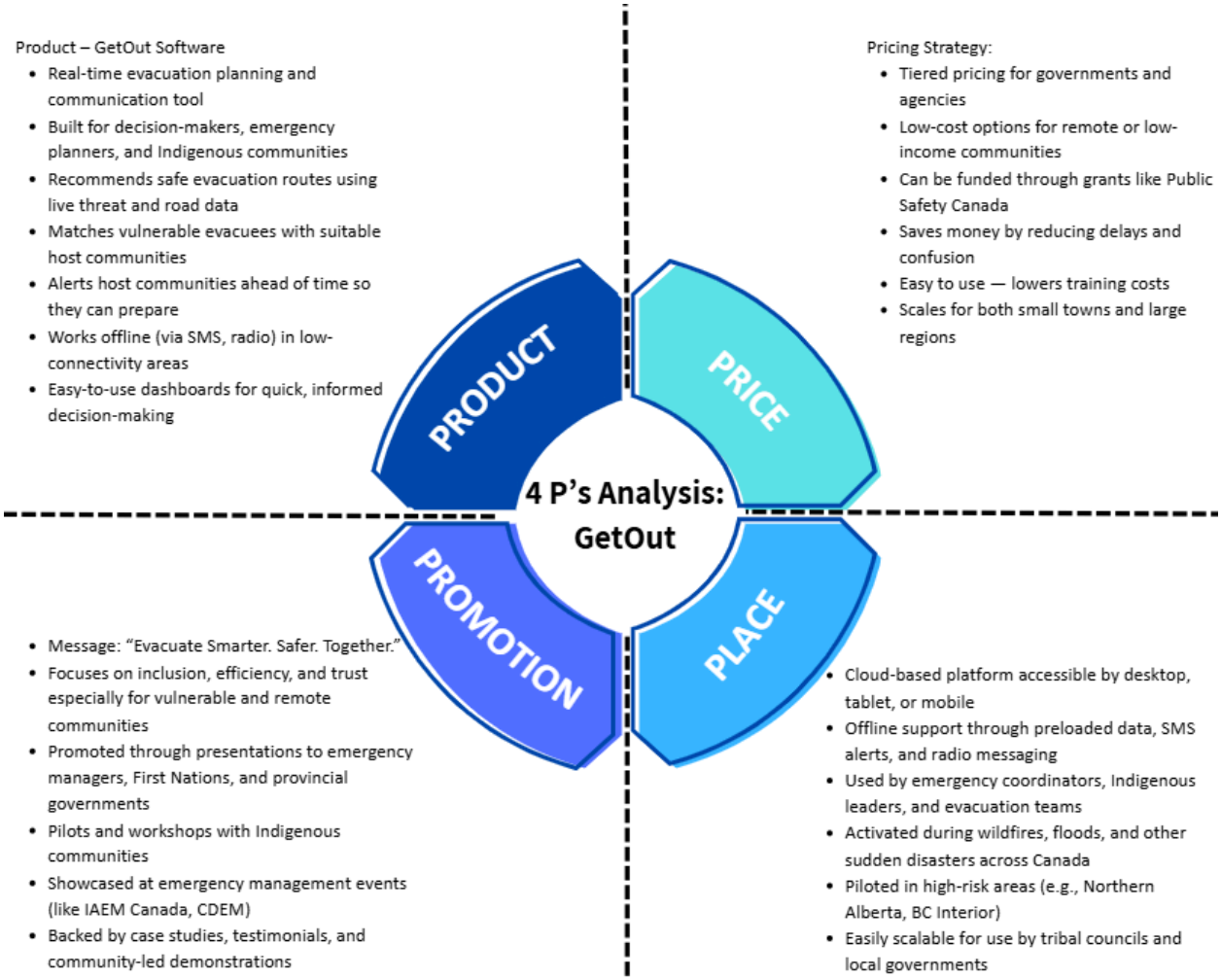
To address the above challenges, the following recommendations may also be considered:

- **Mesh Network for Redundancy**: Use a mesh network to handle situations where communication services are down. Tools like Briar (https://briarproject.org/) achieve this via Wi-Fi and Bluetooth with a regular internet operation mode in case no other devices are nearby. Meshtastic (https://meshtastic.org/) utilizes radio and GPS to establish its mesh network while also allowing access to the network via Wi-Fi, Bluetooth or USB. This method is more reliable, but also requires additional, albeit usually inexpensive equipment.

- **Responsive Design**: Software should work across all devices and screen sizes to ensure usability in various field conditions.
- **Self-Registration Tools**: Systems like ERA (Evacuee Registration and Assistance) allow communities to track households and evacuees in real time, with features such as vehicle tracking and household details.
- **Direct Communication Channels**: Creating a two-way communication method between evacuees and host communities—such as real-time updates, medical info sharing, and community announcements—can reduce stress and improve decision-making.

For more information, see Appendix A for additional recommendations from our interviewees, and Appendix C for specific analysis on existing solutions.

4 Ps & SWOT Analysis

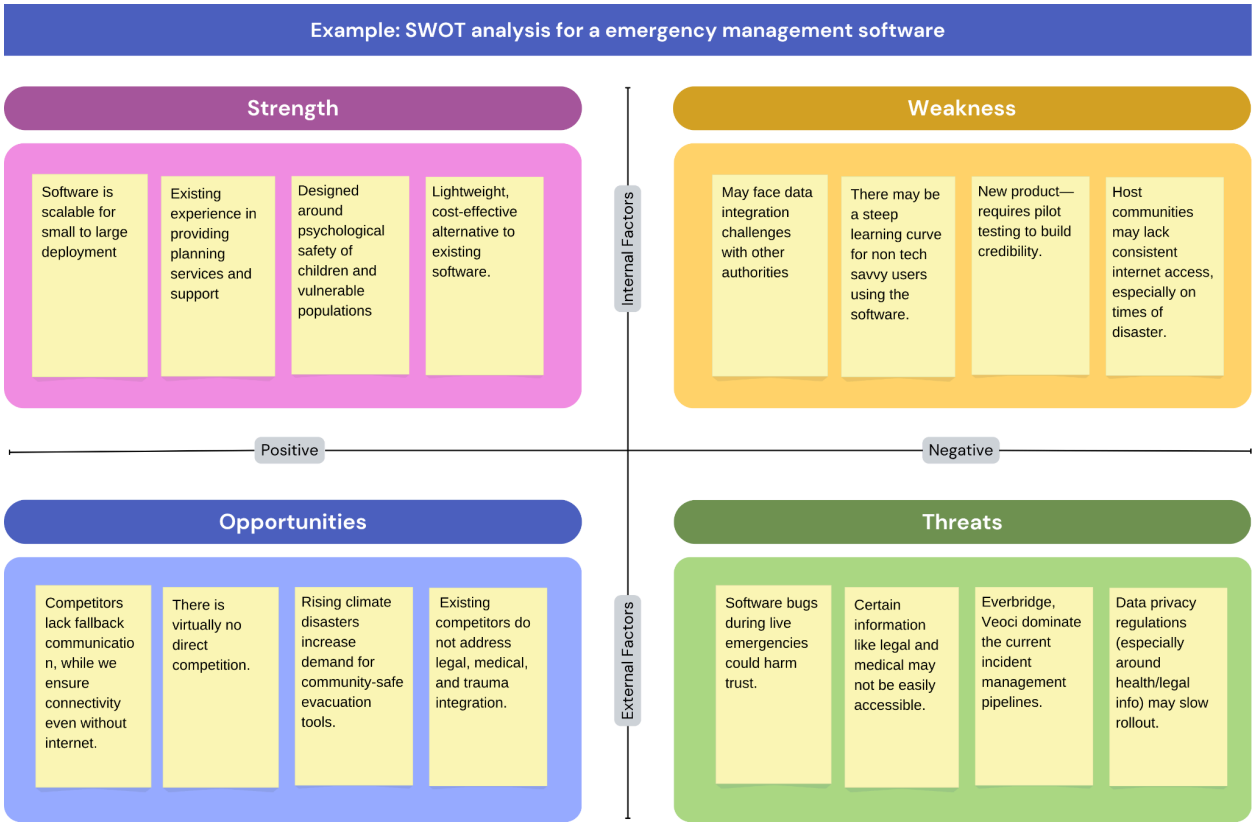


Product
GetOutSoftware is a real-time evacuation planning and communication tool developed to address the unique needs of remote and vulnerable populations. It supports decision-makers and emergency planners by recommending safe evacuation routes, alerting host communities in advance, and functioning even in areas with limited connectivity through mesh networking. Its user interface is designed to support fast, informed, and efficient decisions through simple and easy-to-use dashboards.

Price
The pricing strategy is designed to be inclusive and scalable. Governments and public agencies are the primary clients, supported through a license-based or grant-funded model. For smaller communities or organizations, affordable options and potential subsidies are available. The solution is cost-efficient by reducing confusion during emergencies, lowering training costs, and supporting communities that traditionally lack access to such tools.

Place
As a cloud-based platform, GetOutSoftware is accessible via desktop, mobile, or tablet, and can be activated in diverse emergency scenarios—such as floods or wildfires. It's especially suitable for deployment in Indigenous regions and rural communities across Canada, with field testing planned in high-risk areas like the BC Interior and Northern Alberta. Its infrastructure enables use even without internet access, through preloaded data and alternative communication channels.

Promotion
The product is promoted through a combination of community engagement and institutional outreach. Its message, "Evacuate Smarter. Safer. Together." reflects its core values of inclusivity, preparedness, and local trust. Promotion channels include government presentations, Indigenous-led workshops, and visibility at emergency management conferences. It will be further supported by field-tested pilots, user testimonials, and case studies that will validate its efficacy to the prospective clients.



Strengths
It is scalable, making it suitable for both small-scale and large-scale deployments. The team behind it has substantial experience in providing planning services and support, adding credibility to its implementation. Furthermore, the software is thoughtfully designed with a focus on the psychological safety of children and vulnerable populations, which is crucial during crisis situations. Lastly, its lightweight and cost-effective nature makes it an attractive alternative to existing, often bulkier or more expensive, solutions.

Weaknesses
Despite its strengths, the software also has some internal limitations. One significant concern is the potential difficulty in integrating with systems used by other authorities. Additionally, non-tech-savvy users may encounter a steep learning curve, which could hinder adoption. As a new product, it lacks an established reputation and would require thorough pilot testing to gain trust and credibility. Moreover, in disaster-prone areas, consistent internet access may not be available, limiting the software's functionality in critical times.

Opportunities
There are substantial external opportunities that the software can capitalize on. A major advantage is that many competitors lack fallback communication options, while this software ensures connectivity even without internet access. One way to do it would be mesh communication where devices connect directly to each other, creating a decentralized network where messages can be relayed through multiple paths to reach their destination. The following examples show that these systems have proven themselves over time.In 2011, during the Fukushima nuclear disaster in Japan, mesh networks maintained communication when traditional infrastructure failed.The 2015 Nepal earthquake saw Disaster Tech Lab deploy mesh

networks to assist rescue efforts and provide Internet access to remote communities.In 2018, when a devastating tsunami hit the Indonesian island of Sulawesi, local organizations employed mesh networks to enable vital communication between relief teams and survivors, aiding distribution and rescue operations.

The market is relatively untapped with virtually no direct competition, offering room for growth. Increasing climate-related disasters are raising the demand for tools that prioritize community-safe evacuations. Additionally, existing solutions do not sufficiently address the integration of legal, medical, and trauma support, giving this software a unique edge in addressing those vital needs. GetOutSoftware can pre-register residents from the local government's database and note any special accommodation an individual requires like restraining orders, health conditions and medications and special needs for especially vulnerable people like children and elderly including psychological trauma. After noting these requirements Getout can send requests for these resources which can shorten the time officials might require to coordinate these responses and the response itself would be more effective.

Threats
Several external threats could affect the software's success. Technical failures such as software bugs during live emergencies could damage user trust and reliability. Accessibility of sensitive legal and medical information may be restricted, limiting effectiveness. The market is currently dominated by established players like Everbridge and Veoci, which may make it difficult for a new entrant to gain traction. Finally, stringent data privacy regulations—especially around health and legal information—could delay rollout or complicate compliance, particularly across jurisdictions.

Strengths and Weaknesses Comparison

Strengths	**Weakness**
Designed for remote, vulnerable communities with offline capability.	Still under pilot phase; needs further testing and feature refinement.
Available across devices; user-friendly dashboards.	Internet-dependence in some functions may limit use in extreme situations.
Strong brand message	Limited current awareness

Gaps in the Market and Potential Opportunities

Gaps

While many emergency management software platforms exist—such as EmergencyOS by FacilityOS, Stratus-io, D4H, Navigate360, ArcGIS Emergency Management, Noggin, and The Residential Evac App—most are limited in scope and functionality. These tools typically focus on one specific pillar of emergency response, such as personnel tracking, mapping, or communication. As a result, municipalities and agencies often need to use two or more systems simultaneously to achieve a complete response, leading to inefficiencies, high costs, and operational confusion during critical moments.

Another major shortcoming of existing platforms is their reliance on constant internet connectivity. Many remote and underserved communities, especially in Indigenous regions, do not have reliable internet infrastructure—rendering these tools ineffective when they are needed most. Furthermore, the lack of built-in redundancy (such as SMS or radio-based communication) makes these solutions unreliable in power outages or disaster scenarios where infrastructure is compromised.

Crucially, most current systems do not integrate with the Incident Command System (ICS), a standardized framework used by emergency services across Canada to coordinate response roles, resources, and reporting. The absence of ICS form compatibility creates administrative challenges and hinders smooth coordination between agencies, especially during high-stakes events that require cross-jurisdictional collaboration.

Finally, there is a critical gap in how these systems support vulnerable populations. Evacuation plans often fail to consider medical, and logistical needs, and host communities are rarely given timely information to prepare for incoming evacuees.

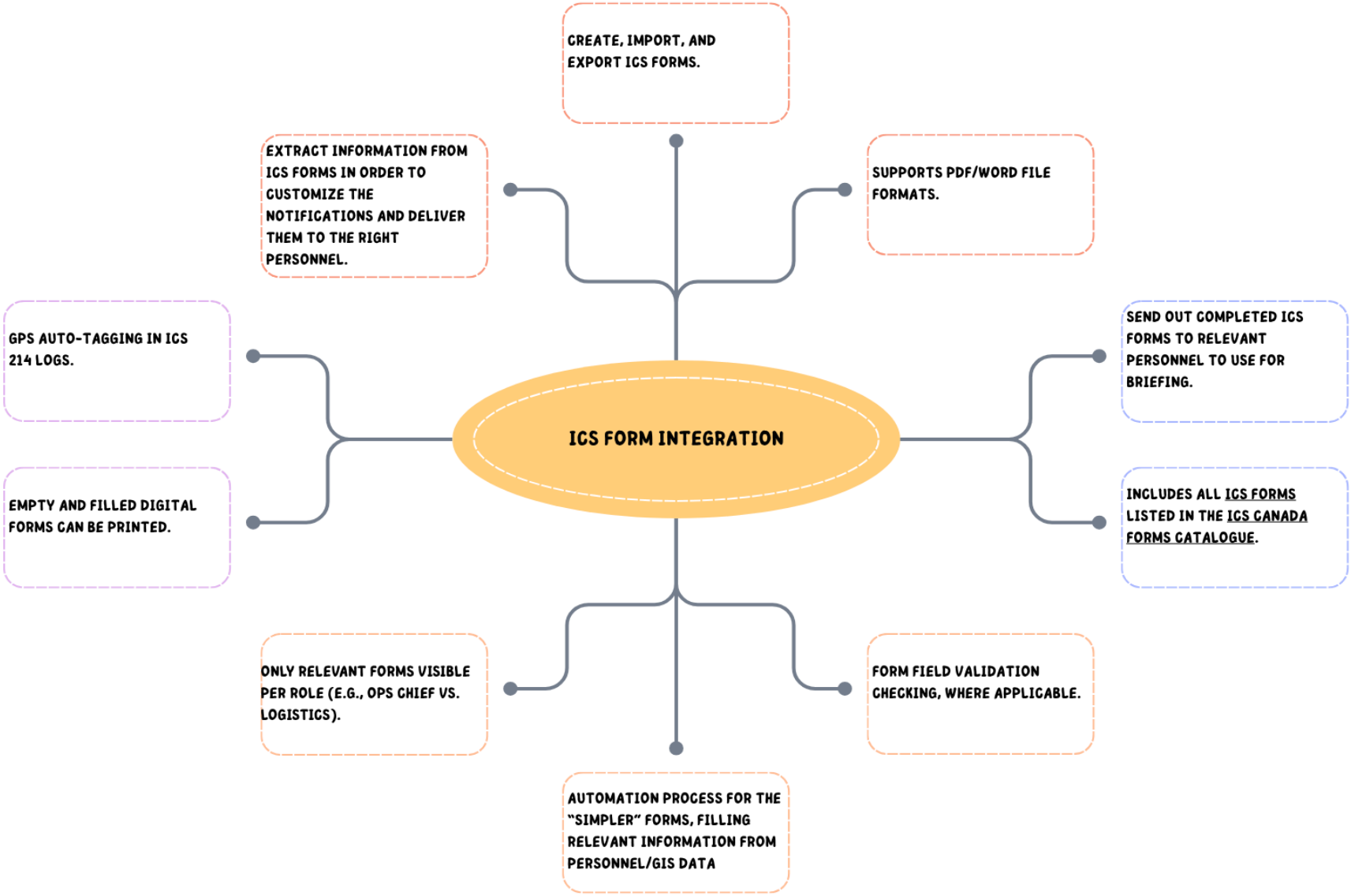
Potential Opportunities

GetOutSoftware is a good fit because it is simple to use, even for people who are not very familiar with technology. The platform will include offline features like mesh networking, which are helpful in areas where the internet is slow or unavailable during emergencies. This makes it possible for decision-makers and responders to stay connected and informed at all times. GetOutSoftware also supports the needs of people who may be at greater risk during disasters—such as the elderly, people with medical conditions, or those with mobility challenges—by helping match them with safe evacuation routes and suitable places to stay. These features make GetOutSoftware not just a tool for emergency planning, but a system that protects lives more effectively when it matters most.

Another important opportunity for GetOutSoftware is its ability to work with ICS forms, which are part of the Incident Command System, a standard method used by emergency services to organize people, resources, and information during a crisis. Many emergency software tools do not support ICS forms properly, which causes delays, confusion, or extra work for responders. GetOutSoftware fills this gap by allowing emergency teams to complete, share, and update ICS forms directly within the platform in real time. This helps everyone stay on the same page, reduces paperwork, and improves coordination between different groups involved in evacuation.

Mind Map and Business Model Canvas

When brainstorming ideas for what the key features for GetOutSoftware would be, the most prominent ones were expanded upon via a mind map, which goes a bit deeper into what such a feature would provide. From there, a couple select mind maps were chosen to go into even more detail as a business model canvas, which lists possible customer segments and relationships, value propositions, channels, and key partners, activities and resources. Our team came to the conclusion that if only one feature had to be developed, it would be integrating ICS forms into a digital, streamlined process. The following two pages detail what that feature would entail. For additional mind maps and business model canvases of our other prominent ideas/features, please refer to Appendix D.


ICS FORM INTEGRATION
CREATE, IMPORT, AND EXPORT ICS FORMS.
SUPPORTS PDF/WORD FILE FORMATS.
SEND OUT COMPLETED ICS FORMS TO RELEVANT PERSONNEL TO USE FOR BRIEFING.
INCLUDES ALL ICS FORMS LISTED IN THE ICS CANADA FORMS CATALOGUE.
FORM FIELD VALIDATION CHECKING, WHERE APPLICABLE.
AUTOMATION PROCESS FOR THE "SIMPLER" FORMS, FILLING RELEVANT INFORMATION FROM PERSONNEL/GIS DATA
ONLY RELEVANT FORMS VISIBLE PER ROLE (E.G., OPS CHIEF VS. LOGISTICS).
EMPTY AND FILLED DIGITAL FORMS CAN BE PRINTED.
GPS AUTO-TAGGING IN ICS 214 LOGS.
EXTRACT INFORMATION FROM ICS FORMS IN ORDER TO CUSTOMIZE THE NOTIFICATIONS AND DELIVER THEM TO THE RIGHT PERSONNEL.

Business Model Canvas

ICS Form Integration

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
• ICS Canada • IT infrastructure partners (Cloud providers) • Training organizations and consultants in emergency management.	• Digital transformation of ICS forms • Automated form filling which fills fields using GIS and personnel data to save time and for error reduction. • Form validation to check entries for accuracy and to prevent mistakes. • Allows PDF/Word export and shows forms based on user roles • GPS tagging which adds location data to event logs and forms automatically • Send completed forms to appropriate personnel • Save forms to local storage • Print out forms	• Reduced manual paperwork and errors • Centralized and secure storage for all ICS documentation • Automation of simple form fields using Geographical Information System (GIS)/personnel data. • Delivering more relevant information based on ICS forms • Streamlined the process of sharing ICS documents	• Continuous improvements through user feedback. • Custom deployment services for complex agencies • Training modules for ICS workflow adoption • Dedicated onboarding and support teams	• Incident Management Team • Fire, police and emergency medical services agencies • Emergency Operations Centers (EOCs). • Non-governmental organizations. • Event organizers

Key Resources	Channels
• Skilled software and mobile application developers • ICS subject matter experts • Secure and scalable cloud infrastructures • UX designers for emergency-friendly interfaces • Integration/API developers	• Web platform for command centers • Mobile application for field personnel • Emergency management training programs • Integration with various organizations' internal systems (eg. GIS, HR databases)

Conclusion

The findings of this consultation make it clear that Canada's current emergency management landscape, particularly in rural, remote, and Indigenous communities, faces persistent gaps in communication, coordination, and inclusivity. Case studies from the High River Flood, Yellowhead County Wildfires and Floods, Lytton Wildfire, and Wood Buffalo Wildfire consistently reveal similar shortcomings: fragmented systems, a lack of real time situational awareness, barriers for vulnerable populations, and slow adoption of effective technologies. These challenges are compounded by reliance on outdated or overly complex tools, creating unnecessary risks in already high pressure situations.

GetOutSoftware emerges as a viable and timely solution to these issues. By integrating robust ICS form functionality, supporting offline operations through SMS, radio, or mesh networks, and prioritizing culturally responsive and accessible design, the platform directly addresses the operational and human needs identified in both primary and secondary research. Its modular, intuitive approach reduces complexity while enabling interoperability, which is key to fostering trust and adoption in communities that have historically been underserved by emergency technologies.

Moving forward, the development of GetOutSoftware should remain grounded in collaboration with frontline responders, community leaders, and end users. Continued field testing, refinement based on real world feedback, and a focus on interoperability with existing emergency frameworks will be critical to its success. By aligning technological innovation with practical usability, GetOutSoftware can become more than a tool, it can be a bridge between agencies, communities, and the resources they need to respond effectively to disasters.

In a climate where extreme weather events and large scale emergencies are becoming more frequent, the urgency for reliable, inclusive, and adaptable solutions has never been greater. If implemented as envisioned, GetOutSoftware has the potential to transform emergency management in Canada, enhancing resilience, protecting lives, and empowering communities to face future crises with confidence.

Appendix A - Interviewee Notes

As mentioned in our *Key Findings and Insights* section, the following notes that were taken from the interviews show that there were issues with communication and planning for emergencies. In addition, there's a consistent lack of technology adoption among rural and remote communities. These findings are what drove our team to choose ICS form integration as our most prominent idea/feature for GetOutSoftware.

Vanessa Law

Personal Background

- Works for Daylu Dena Council,a Kaska community in Lower Post, a northern community in B.C.
- Was the former Lands Director.
- Presently, is doing a one-year position as an Emergency Program Coordinator.
- Previously, he has been the organization's Emergency Operations Centre Director. (off the desk).
- Has coordinated evacuations during 2018 wildfires, and set up EOC during flood risk and COVID.

Overview of the community and technology used

- Approximately 160 residents and 75 homes.
- Very remote, limited cell service calls are made over Wi-Fi.
- Most residents use Starlink for the internet.
- A high-speed fiber internet line is being planned to service the main office.

Questions

- **Primary mode of communication during emergencies?**
  Landlines and cell phones. The majority have Starlink as their internet provider. Currently, the community is planning to install a high speed, high fiber internet line into the community to at least connect to their main office.

- **Main challenges communicating evacuation orders to communities and families?**
  Not many challenges observed.

  In 2018, communication was through landlines or cell phones. Facebook notifications were also used.

  Door-door notices were used for people who did not have phones. The problem with this is that it may be hard to manage time during emergencies. Example: Evacuation drill in the previous year involved 3 wildfire crews (teams of 2); Took them 2 hours to evacuate completely.

  Competitors: Hawk Solutions

- **How do you tailor evacuation information to affected children and youth?**
  Nothing is tailored any differently.

- **What emergency/crisis management software solutions are you currently using?**
  Nothing. Briefly looked into Hawk Solutions.
- **What 3 features would be most important for you in regards to emergency/crisis management software?**
  Should be easy-to-use. self-registration feature for evacuees recommended.

  Her community is in BC, next to the Yukon border. As a first Nation, their funding comes from Yukon. A lot of the communities in Yukon lack emergency planning data.

  In Vanessa's community, she uses a tenant list that includes all the house numbers, the number of people in each household, which households have vehicles,etc. It can be updated accordingly. They have a tool called ERA (Evacuee Registration and Assistance) that helps evacuees self-register their information for support during evacuations.
- **How do people without smartphones or access to the internet still get important updates about the evacuation?**
  Door to door, landlines, publicly viewable Billboard.

James Robicheau

Interview Questions

- **Could you tell us a bit of your background?**
  Started in the oil rig industry in 2003. Currently residing at High Level from 2006. Learn more about supply chain management. He has a diploma in Addiction Counselling.
- **What are the main challenges you've seen in communicating evacuation orders to communities and families?**
  First nations tend to not want to leave their lands or far from it, during a state of local emergency.
- **What emergency/crisis management software solutions are you currently using?**
  Excel is being used to manage inventory expiry dates.
- **Are there any current pain points with the software solutions you use?**
  Requires a lot of manpower to properly manage the inventory auditing.
- **What 3 features would be most important for you in regard to emergency/crisis management software?**
  A system that monitors expiry dates. Inventory Control system that adjusts the inventory count.
- **What's the primary mode of communication (Internet, SMS, Radio, etc.) for incidents or emergencies, and what happens if that method becomes inaccessible?**
  ICS sheets/forms to communicate. The primary mode of communication used is SMS.

James mentioned that there are people who get penalized for going above and beyond for rural communities. Essentially becoming black listed.

Other Highlights

- He uses spreadsheets to track expiry dates for his inventory. He mentioned that he forgets to throw things out sometimes.
- He uses pen and paper documents to keep track of what's taken from the inventory.
- He mentioned "SAP" for inventory management, that it's "too much", so he doesn't need it.
- They don't have any communication in cases when evacuees need special medications. So they don't prepare in advance. But he mentioned that if something like this happens, they can just call the doctor and they have delivery ready to go.

Leslie Ann

Interview Questions

- **Could you tell us a bit of your background?**
  Life long public servant. Wildfire information officer at Alberta Forestry. Created and maintained relations with stakeholders in wildfires in Alberta. 2010 Wildfire dispatcher. 2016 Fort McMurray wildfires caused Leslie Ann to struggle as it affected her physically. Currently employed with the City of Edmonton.

- **What are the main challenges you've seen in communicating evacuation orders to communities and families?**
  The biggest challenge is being able to leverage social media because of misinformation and rumors.

- **How do you tailor evacuation information to affected children and youth?**
  Leslie said she has not seen any information tailored to children.

- **What emergency/crisis management software solutions are you currently using?**
  ICAD is GPS tracking.
  - Can you give us an example of how such software is used in a real incident?
  - Are there any current pain points with the software solutions you use?

- **What 3 features would be most important for you in regards to emergency/crisis management software?**
  Real time information of various resources such as hotels and supplies.

  Mapping escape routes for evacuees.

  Pushing a list of proper items to take with you during an evacuation. Preparing one's property for an evacuation.

- **What's the primary mode of communication (Internet, SMS, Radio, etc.) for incidents or emergencies, and what happens if that method becomes inaccessible?**
  Radio/wifi is the primary mode of communication.

The AFRAC system is designed to divert overloads/excess to other centres with excess capacity.

Shari Olsen

Software Mentioned

Alertable (she likes it)

- able to customize what alerts to receive
- able to customize the method to receive the alerts

Issues for the Evacuees

- Not knowing where to go
- Not knowing what to bring
- Not knowing what to do
- No cell phone service or internet access sometimes

Top 3 Features

- Accessibility
- Use plain language
- Tells the user what to do

Modes of Communication

- Phone
- In person/face-to-face

Paul Harris

Personal Background

- Worked in risk assessment, planning, training
- Worked with both singular command and unified command structures
- Had experience with many software solutions before

Software in Use / Have Used Before

- Mostly manual document work
- Microsoft Onedrive
- IAP by The Response Group
    - Requires too much training to use the software
    - Difficult to operate. Usually takes a whole team
- D4H
    - Too many features
- WebEOC (not used but considered)
- H2Safety (or any GIS based solutions)

    - Too complex, hard for them to understand. Sometimes it’s better to just use Google Maps.
    - Steep learning curve. People shouldn't have another thing to worry about when they are under the stress of the incident already.

For Communication

- Mostly verbal, face-to face
- Telephone and text messages
    - If service is down, there would just be no response
- WhatsApp groups
- Microsoft Teams
- Microsoft Word for documents
- Email

Other Issue Faced

For the evacuees: They usually don't know what to do until the incident happens. And when it happens, mixed information is all over the internet, which creates uncertainty and fear.

Feature Preferences

- He thinks software can be useful for giving shelters, less so for evacuation
- He prefers simple software for simple incidents, because they tend to not require complex features
- Emphasis on the first 4 / 24 / 48 hours
- Easy sharing of actionable intelligence, instead of information dump like some ICS forms do (used ICS201 as an example)

Appendix B - Additional Information on the May 2016 Wood Buffalo Wildfire

Lessons Learned Prior to the Wildfire

- Formalized Public Safety Governance
- Prevention programming
- Enhanced systems of support for local authorities
- Enhanced emergency management training and exercises
- Mandated provincial use of the Incident Command System model
- Improved Provincial Operations Centre capacity and technology
- Formalized Incident Management Teams
- Launch of the Alberta First Responder Radio Communications System
- Implementation of the Alberta Emergency Alert System
- Improved integration of provincial expertise for environmental and public health review prior to re-entry
- Planning for re-entry and recovery

Lessons Successfully Applied to the Wildfire

- Evacuation of the Urban Service Area of Fort McMurray took only a day. Registration Centres were promptly set up across Alberta to support affected residents, providing shelter and social services.
- Firefighters protected critical infrastructure and minimized damage to the Urban Service Area, allowing quicker re-entry/recovery efforts.
- The Telephone Town Halls were established to connect the Region's displaced residents together. They were set up and managed by the Premier's Office and Public Affairs Bureau respectively, and it included a community forum to provide Q&A for affected residents, which in turn gave insight to said residents' needs.
- The Rapid Damage Assessments was developed and leveraged during the Region's stabilization. It assessed the damage of all buildings caught in the Wildfire and displayed the information via a dedicated website, helping residents plan for their return to the Region after the Wildfire subsided.
- $319 million was donated to the Canadian Red Cross as a result of this catastrophe.
- Based on these lessons learned, the Province of Alberta strived to:
    - Increase investment in emergency prevention programs
    - Improve public emergency awareness and preparedness
    - Continue supporting local authorities via incident response and evacuation protocols
    - Enhance technology and analytics usage in emergency management processes
    - Further operationalize the delivery of emergency social supports

Recommended Lessons to Learn from the Wildfire

There was confusion over who actually had the authority to allow re-entry into the Region (i.e. one of the ministers that were relevant to the situation, Alberta Environment and Parks, etc.). As such, it was recommended to re-analyze the following Acts: Emergency Management Act, Public Health Act, Environmental Protection and Enhancement Act, Occupational Health and Safety Act.

Local and provincial authorities should continue to invest in disaster prevention programs. The amount spent on such programs in 2015/2016 was approximately $53 million, which was a mere 0.5% of the cost of the disaster. This ongoing investment, as well as cost avoidance methods, are absolutely needed for the Government of Alberta to grow not just disaster prevention, but overall emergency management.

Guidelines should be made to help support local authorities in creating a disaster resiliency strategy. One suggestion would be to refer to the United Nations' Sendai Framework for Disaster Risk Reduction. This framework prioritizes four things for disaster risk management. First, understand disaster risk through data collection, developing baselines, information sharing, and government collaboration. Second, strengthen disaster risk governance via plans, guidance, and coordination with stakeholders. Third, invest in disaster risk reduction/prevention to increase communities' overall resilience to such disasters. This includes promoting disaster risk assessments at the local level. Finally, enhance disaster preparedness and "Build Back Better" by periodically reviewing/updating existing disaster preparedness plans, maintaining investments in disaster forecasting/monitoring/warning systems, promoting both resilience in infrastructure and public awareness for emergency preparedness, and preparing workforce participation in emergency responses.

Development for an internal Emergency Management Communications Interoperability Plan is highly recommended, one which aligns with the Communications Interoperability Strategy for Canada. This plan should include all levels of government, Indigenous communities, RCMP and police, industry, service providers, and other emergency response partners. The plan itself should include a more formal engagement approach for points of contact between the Government of Alberta, local authorities and industry stakeholders. In addition, the plan must have appropriate policies and processes for cellphones, so that communication channels remain available during an incident. This does not mean to reduce other methods of communication; in fact, other forms of emergency communication should also remain in use, such as the Alberta First Responder Radio Communications System and the Alberta DisasterLAN. Finally, this plan requires Internal and Public Information Officers to support the flow of information within the responding organizations and stakeholders, as well as the general public.

Existing public emergency communication approaches should not just continue to be utilized, but also enhanced. The Province of Alberta should make emergency communications more accessible for the hearing impaired, deliver information booklets containing important information during re-entry, and enhance leveraging social media in emergency situations. It was found that the primary sources of information during the evacuation were radio and social media, the latter being specifically Twitter and Facebook. Updates were posted on a regular basis via the Region's and Mayor's accounts. Video updates and live feeds were also uploaded to YouTube. This flow of information made it rather difficult for the affected residents and general public to properly receive.

The execution of Rapid Damage Assessment technology and processes during the Wildfire was a net positive and thus should continue to not only be used, but expanded upon. One example of such technology was satellite imagery, which allowed Safety Code Officers to assess the scale and scope of the structural damage caused by the Wildfire. The images captured by the satellite, as well as the relevant information pertaining to it, were then made available to affected residents via a website. By allowing the residents to also search and access the status of their homes, they could make the appropriate preparations for both their return and their futures. In order to leverage the success of Rapid Damage Assessments for future incidents, local authorities would need provincial support by way of geographic information system capabilities and mapping.

Additional Recommendations

- Standardize the format and processes for post-incident assessments
- Enhance investment in public awareness and engagement initiatives for emergency preparedness
- Clarify and document processes for legislative delegation of authority in the Alberta Emergency Plan
- Scale the Hazard Identification Risk Assessment model to understand cumulative risk
- Create resource and asset management system capabilities
- Develop a Provincial Emergency Evacuation Framework and evacuation model to provide enhanced decision-making capabilities at the Provincial level
- Build depth and capacity within local authorities to enable communities to support each other during emergencies
- Invest in and develop a state of the art Provincial Operations Centre facility

- Mandate local authorities to adopt the Incident Command System model during response
- Formalize criteria and operational processes for de-escalation of the Provincial Operations Centre
- Empower and resource the Office of the Fire Commissioner to coordinate and deploy municipal firefighters during a response
- Continue to operationalize the Provincial Emergency Social Services Framework
- Build re-entry conditions and criteria into a guide that can be leveraged by local authorities
- Develop recovery plans within the context of resiliency
- Enhance the use of the AEMA Recovery Branch

Appendix C - Detailed Analysis on Existing Software and Non-Software Solutions

Everbridge

- Pricing is restrictive
- Although User interface is modern but training of users can be time-consuming
- No support for evacuee-level needs (e.g., medication, legal status, trauma screening).
- Great for communication, but lacks tools for host community logistics or resource planning.
- Has diversified market base such that emergency management is just a piece of its operations

DisasterLAN (DLAN)

- DLAN is the only Emergency Management Software to have completed full NIMS-STEP testing
- DLAN provides many tools for daily operations including: work order management; event logging; social media monitoring; email monitoring; news and weather monitoring; document sharing; user and role activity logging; and communication continuity.
- Users require at least ICS 400 certification so Steep learning curve for *New Users*
- User interface feels outdated
- Cannot flag individual evacuee needs like medications or restraining orders.
- Price is restrictive to small communities

D4H

- Arguably the most popular crisis/emergency management software currently available.
- Mobile and web applications.
- Software is modular, meaning you pay and use what you want. This means separate pricing plans for each module/product, separated as such:
    - Incident Management
    - Alerting
    - Personnel & Training
    - Equipment Management
    - Incident Reporting

    - Cloud Plans (always required)
- Pricing for each module is separated into four different plans: basic, advanced (most popular), professional, and enterprise. There is, however, a 25% annual cloud plan that's required for all customers. Lower plans of course have a more limited amount of features, as well as number of users/recipients that are able to access the software.
- A grant can be given by the company for small teams (80 people or less) consisting entirely of unpaid volunteers.
- Highly favorable reviews in general. Most common criticism seems to be that any features suggested by users/customers take a while to implement.

GSAR (Ground Search and Rescue)

What it does:

- GSAR teams (nearly 300 teams and 9000 volunteers) help locate missing persons in wilderness or disaster areas 24/7, 365 days a year, providing their services at no cost to the public.
- They use GPS, GIS mapping, and coordination tools to plan search routes and communicate with command centres.
- Integrates CounterCrisis Tech's IC4W application and the Android Team Awareness Kit (ATAK).

Limitations:

- Mostly volunteer-based . Hence limited resources and training across regions.
- Not fully integrated with national emergency platforms . There is real-time data integration but limited. (confirmed by the GSAR ICS Project, backed by Public Safety Canada).
- General lack of automation for data collection and entry, resulting in insufficient data for knowledge management and policy informs.
- Some existing technologies are reported as difficult to use and unreliable.

When it was used:

- Frequently deployed during natural disasters like **the 2013 Alberta floods** and **the Fort McMurray wildfire in 2016**, where large numbers of residents had to be evacuated and missing persons needed to be located.
- Also used in smaller-scale rural or wilderness rescue operations across Canada year-round.

Who uses it:

- Provincial Emergency Management Offices (For example EMBC in British Columbia), RCMP, and local municipalities.

Juvare (WebEOC & Crisis Track)

What it does:

- Provides centralized dashboards to track incidents, manage resources, and coordinate between agencies.
- Crisis Track is used in the field to collect real-time data, assess damage, and sync with command centres.
- EMTrack ensures secure and scalable tracking of patients and populations during emergencies: uses physical triage tags, barcoded bracelets, or tracking numbers, is HIPAA compliant.

Limitations:

- It has a complex interface: Not very user friendly for first time users. The model needs a lot of training
- Customizing it for different agencies takes time and vendor support.
- Limited interoperability across systems unless standardized.

When it was used:

- Widely used during the COVID-19 pandemic to coordinate hospital capacities, PPE inventory, and vaccination logistics.
- Also activated in British Columbia during the 2021 wildfires and floods, helping track evacuation orders, road closures, and resource deployment.
- Crisis Track was used in the field after tornadoes in Ontario and flooding in Quebec to conduct rapid damage assessments.

Who uses it:

Ministries of Health, provincial EOCs, municipalities, hospitals, and utilities

Navigate360

What it does:

- comprehensive student success and emergency management platform, primarily used in K–12 schools and higher education institutions. It connects recruitment, student retention, mental health, and safety planning in one system.
- 360° Student Profiles with academic progress, alerts, and support history
- Communication Tools for outreach, appointment scheduling, and targeted messaging
- Case Management for tracking academic, behavioral, or wellness-related concerns
- Emergency Operations Plans (EOP) and drill management
- Risk Assessment & Site Mapping tools
- Panic Alerts, live chat, and 911 dispatch integration

Limitations:

- Users report the platform can be initially overwhelming due to its broad feature set, requiring substantial onboarding and customization.
- Navigate360's pricing model may limit access for smaller schools or budget-constrained districts, especially when adding multiple modules.
- Some reviews cite delayed customer service responses or limited post-sale technical support.

When it was used:

- While not used in large-scale natural disasters, Navigate360 is routinely used in school emergencies and planned drills, such as:
  Lockdowns and shelter-in-place events, Active shooter or threat response simulations, Health crisis protocols (COVID-19 contact tracing and isolation tracking),Safety assessments during sporting events, campus festivals, and parent nights

No publicly documented deployments during major natural disasters like wildfires or floods.

Who uses it:

- Over 35,000 schools across North America, more than 5,000 law enforcement and emergency agencies
- Higher education institutions (through the EAB/Navigate360 CRM platform for recruitment, retention, and student support)
- Common users include school principals and safety officers, Campus emergency response teams,IT and facilities staff managing alerts, drills, and mapping

Preppr

https://www.preppr.ai/
An AI powered platform for designing disaster exercises

People like that it comes up with scenarios they wouldn't have thought of otherwise, but are concerned about data privacy, since Preppr is using third party AI providers such as OpenAI.

Non-EMS Solutions

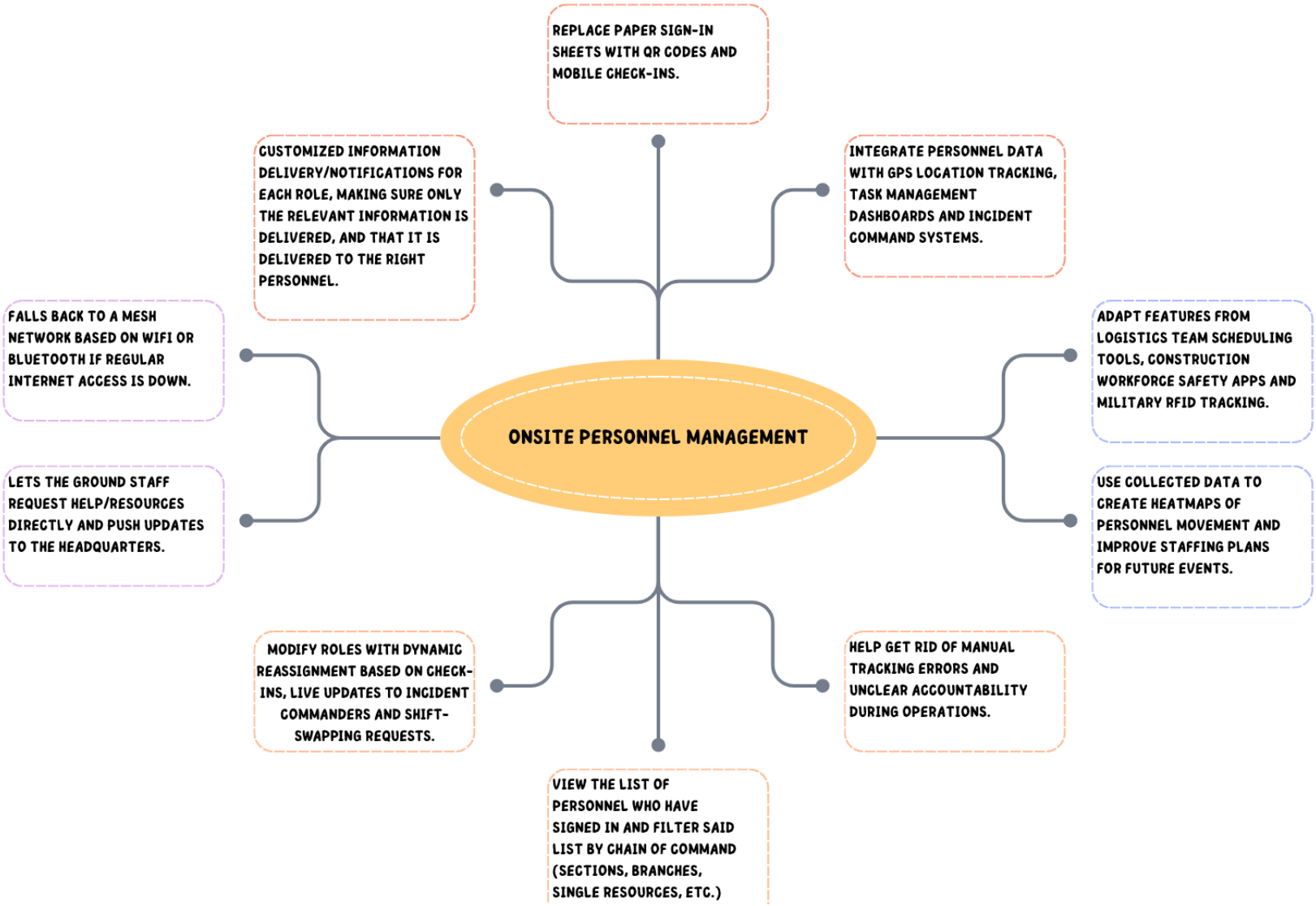
Many organizations are using other software and non-software solutions for emergency management:

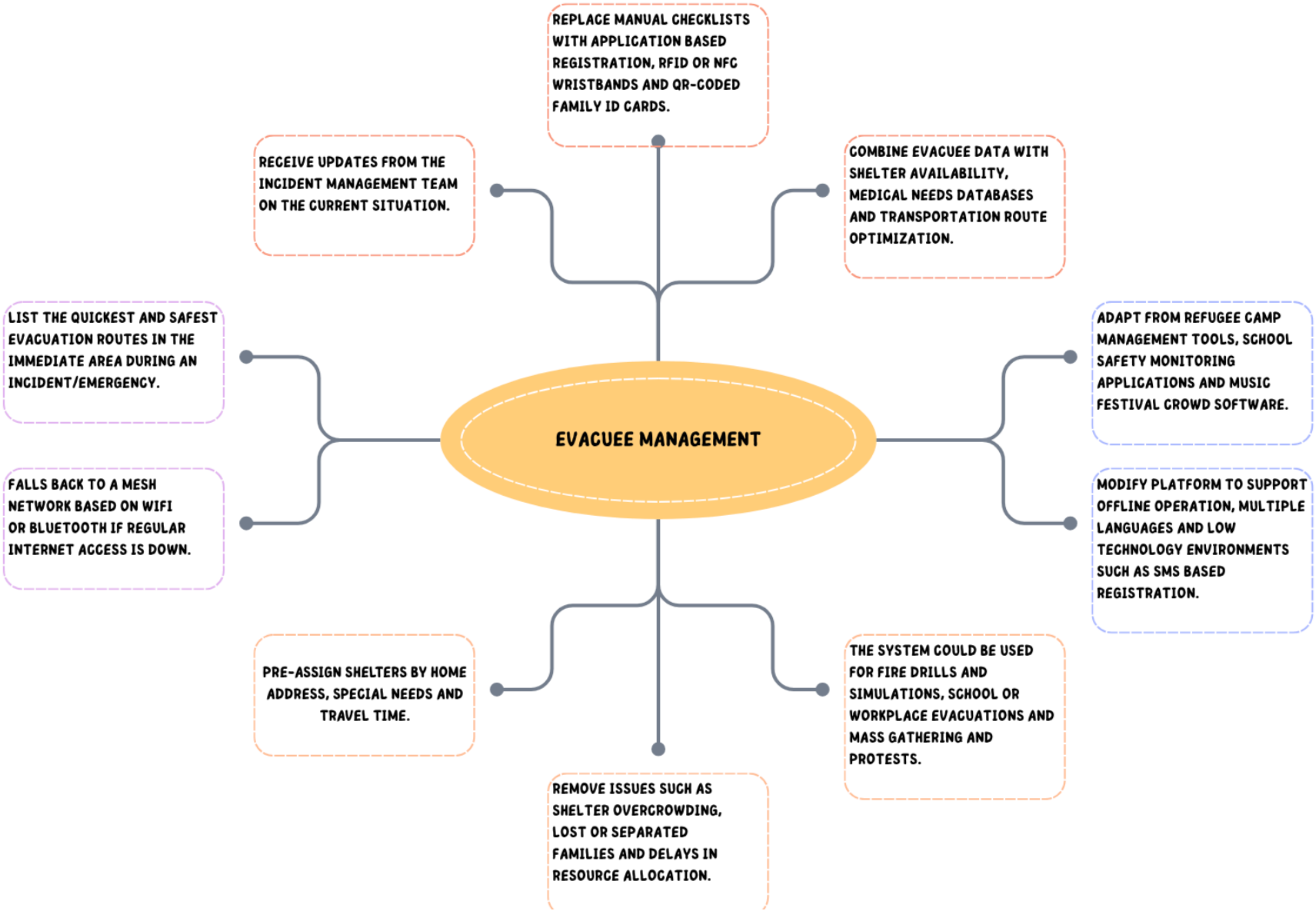
1. Pen & Paper
2. Office suites
    a. MS Project / Visio for planning
    b. MS Teams / Slack for communication
    c. Excel / Google Sheets for data reporting
3. Project Management Software
    a. Jira

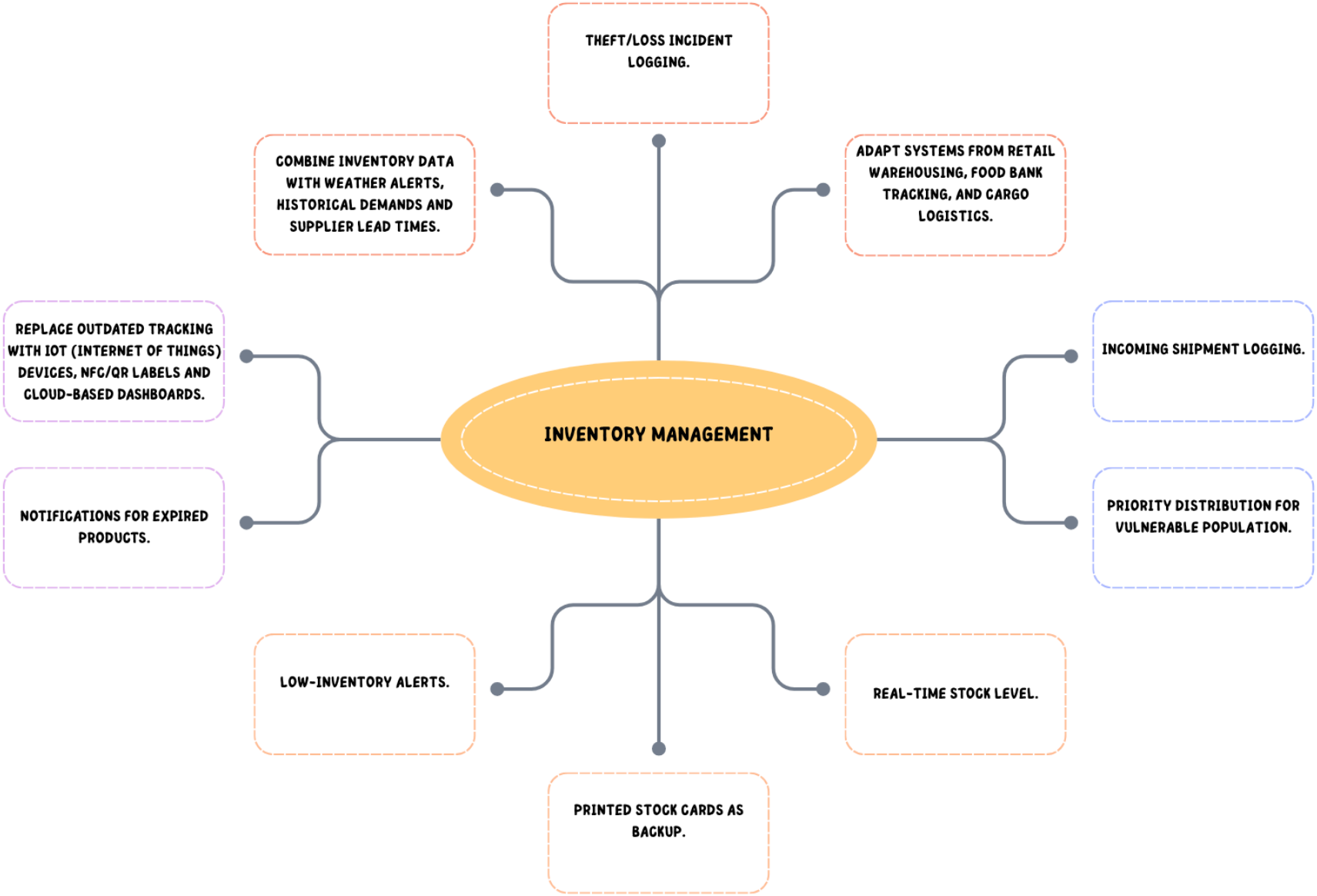
    b. Confluence
4. GIS software to visualize and analyze geographically specific data
    a. ArcGIS Survey123

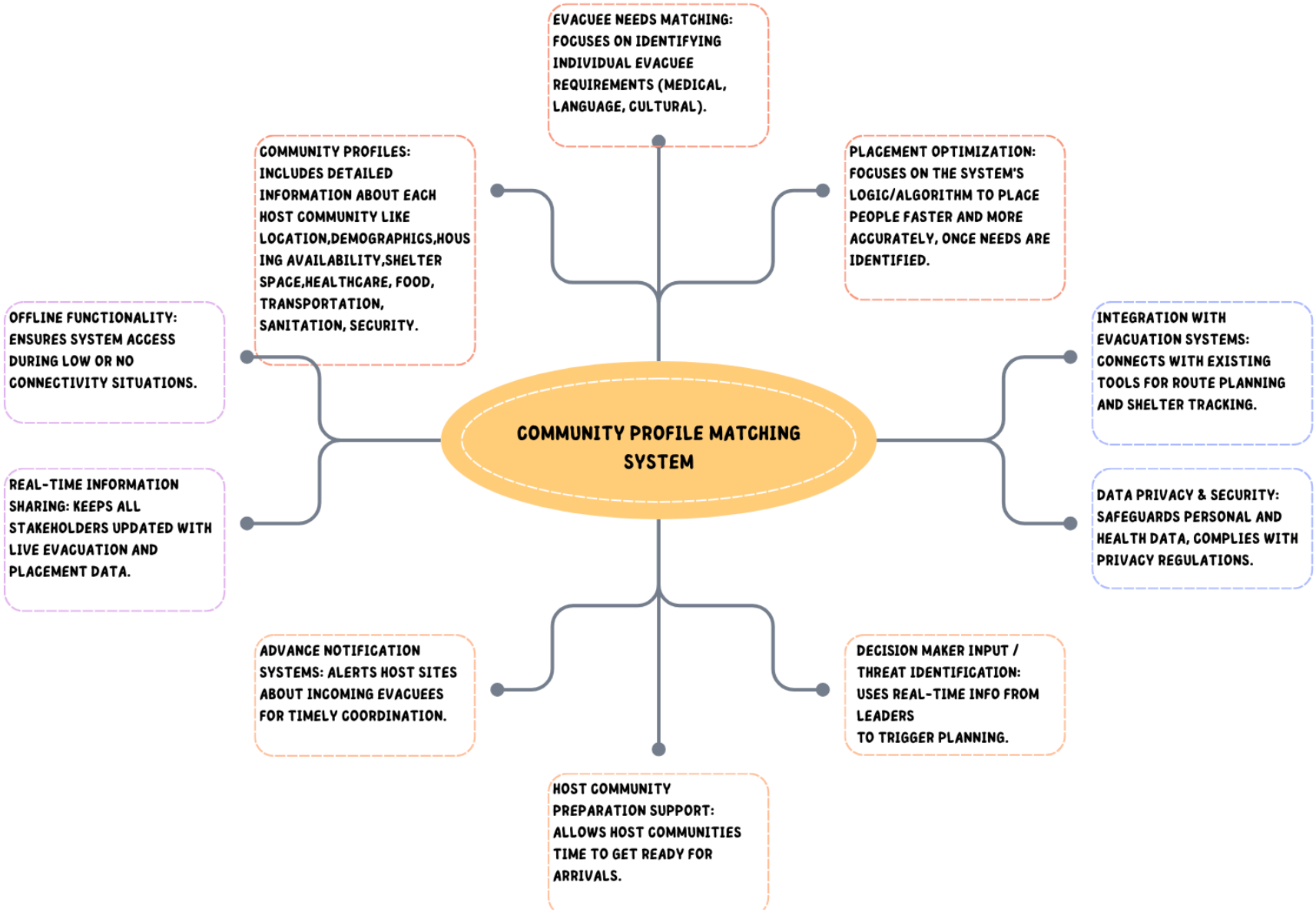
Some of them voiced their frustration with the complexity, lack of flexibility and outdated solutions in the EMS field. They'd rather stick to something more traditional like pen & paper, spreadsheets, PDFs. Or they find something more modern and more customizable, despite the fact that these solutions are not made for emergency management.

Appendix D - Additional Mind Maps and Business Model Canvases




EVACUEE MANAGEMENT
REPLACE MANUAL CHECKLISTS WITH APPLICATION BASED REGISTRATION, RFID OR NFC WRISTBANDS AND QR-CODED FAMILY ID CARDS.
RECEIVE UPDATES FROM THE INCIDENT MANAGEMENT TEAM ON THE CURRENT SITUATION.
COMBINE EVACUEE DATA WITH SHELTER AVAILABILITY, MEDICAL NEEDS DATABASES AND TRANSPORTATION ROUTE OPTIMIZATION.
LIST THE QUICKEST AND SAFEST EVACUATION ROUTES IN THE IMMEDIATE AREA DURING AN INCIDENT/EMERGENCY.
FALLS BACK TO A MESH NETWORK BASED ON WIFI OR BLUETOOTH IF REGULAR INTERNET ACCESS IS DOWN.
ADAPT FROM REFUGEE CAMP MANAGEMENT TOOLS, SCHOOL SAFETY MONITORING APPLICATIONS AND MUSIC FESTIVAL CROWD SOFTWARE.
MODIFY PLATFORM TO SUPPORT OFFLINE OPERATION, MULTIPLE LANGUAGES AND LOW TECHNOLOGY ENVIRONMENTS SUCH AS SMS BASED REGISTRATION.
PRE-ASSIGN SHELTERS BY HOME ADDRESS, SPECIAL NEEDS AND TRAVEL TIME.
THE SYSTEM COULD BE USED FOR FIRE DRILLS AND SIMULATIONS, SCHOOL OR WORKPLACE EVACUATIONS AND MASS GATHERING AND PROTESTS.
REMOVE ISSUES SUCH AS SHELTER OVERCROWDING, LOST OR SEPARATED FAMILIES AND DELAYS IN RESOURCE ALLOCATION.


INVENTORY MANAGEMENT
THEFT/LOSS INCIDENT LOGGING.
COMBINE INVENTORY DATA WITH WEATHER ALERTS, HISTORICAL DEMANDS AND SUPPLIER LEAD TIMES.
ADAPT SYSTEMS FROM RETAIL WAREHOUSING, FOOD BANK TRACKING, AND CARGO LOGISTICS.
REPLACE OUTDATED TRACKING WITH IOT (INTERNET OF THINGS) DEVICES, NFC/QR LABELS AND CLOUD-BASED DASHBOARDS.
NOTIFICATIONS FOR EXPIRED PRODUCTS.
INCOMING SHIPMENT LOGGING.
PRIORITY DISTRIBUTION FOR VULNERABLE POPULATION.
LOW-INVENTORY ALERTS.
REAL-TIME STOCK LEVEL.
PRINTED STOCK CARDS AS BACKUP.


COMMUNITY PROFILE MATCHING SYSTEM
EVACUEE NEEDS MATCHING: FOCUSES ON IDENTIFYING INDIVIDUAL EVACUEE REQUIREMENTS (MEDICAL, LANGUAGE, CULTURAL).
COMMUNITY PROFILES: INCLUDES DETAILED INFORMATION ABOUT EACH HOST COMMUNITY LIKE LOCATION,DEMOGRAPHICS,HOUSING AVAILABILITY,SHELTER SPACE,HEALTHCARE, FOOD, TRANSPORTATION, SANITATION, SECURITY.
PLACEMENT OPTIMIZATION: FOCUSES ON THE SYSTEM'S LOGIC/ALGORITHM TO PLACE PEOPLE FASTER AND MORE ACCURATELY, ONCE NEEDS ARE IDENTIFIED.
OFFLINE FUNCTIONALITY: ENSURES SYSTEM ACCESS DURING LOW OR NO CONNECTIVITY SITUATIONS.
INTEGRATION WITH EVACUATION SYSTEMS: CONNECTS WITH EXISTING TOOLS FOR ROUTE PLANNING AND SHELTER TRACKING.
REAL-TIME INFORMATION SHARING: KEEPS ALL STAKEHOLDERS UPDATED WITH LIVE EVACUATION AND PLACEMENT DATA.
DATA PRIVACY & SECURITY: SAFEGUARDS PERSONAL AND HEALTH DATA, COMPLIES WITH PRIVACY REGULATIONS.
ADVANCE NOTIFICATION SYSTEMS: ALERTS HOST SITES ABOUT INCOMING EVACUEES FOR TIMELY COORDINATION.
DECISION MAKER INPUT / THREAT IDENTIFICATION: USES REAL-TIME INFO FROM LEADERS TO TRIGGER PLANNING.
HOST COMMUNITY PREPARATION SUPPORT: ALLOWS HOST COMMUNITIES TIME TO GET READY FOR ARRIVALS.

Business Model Canvas

Evacuee Management

Key Partners

- Government agencies
- Hospitals
- Temporary Shelters
- Fire Chiefs
- Internet Service Provider
- API provider

Key Activities

- Digital registration and evacuee tracking
- Data integration between shelters, health, transport, etc.
- Offline access functionality (mesh network, SMS)
- Enables use in multiple languages and ensures access for people with disabilities
- Adaptable for both emergency drills and real incidents

Key Resources

- Cloud infrastructure (AWS, Azure)
- Software and database development team
- Privacy framework
- Interactive map API

Value Proposition

- Faster and more organized shelter allocation
- Personalized care through medical data integration.
- Improved situationa awareness for responders

Customer Relationships

- Dedicated support during emergencies
- Customizable dashboards for various agencies
- Ongoing training and systems updates

Channels

- Integration with existing emergency systems (APIs)
- Partnerships with local government
- Workshops and training programs

Customer Segments

- Incident Management Team
- Disaster relief Non Governmental Organizations (eg. Red Cross, UNHCR)
- Hospitals
- Temporary Shelters
- School boards for student evacuation protocols.

Business Model Canvas

Community Profile Matching System

Key Partners

- Host communities
- Indigenous leaders and councils
- Data providers
- Evacuation coordinators
- Local police departments

Key Activities

- Offline access functionality (mesh network, SMS)
- Maintain detailed host community profiles(services,housing,culture,etc.)
- Identify and match evacuee needs like medical,language,cultural and levels of vulnerability
- Use algorithms to optimize placements for speed,accuracy and fit according to evacuee needs
- Integrate with evacuation tools like route planners and shelter databases.
- Enable host communities to directly communicate with evacuees

Key Resources

- Route planning software
- AI/ML algorithms for matching

Value Proposition

- Community information is in detailed, yet easy to read format
- Evacuees are matched based on individual needs
- Host communities are notified in advance of incoming evacuees
- Personal information is kept private and secure
- Key partners are kept up to date with evacuation status and placement
- Evacuees kept safe with families and separate from any abusers

Customer Relationships

- Pre-disaster profiling and planning

Channels

- Partnerships with local communities
- Government procurement contracts

Customer Segments

- Incident Management Team
- Municipal government agencies

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